# Performance Optimisation Hands-on Session

You are supplied with a short C program in the file **`perfopt.c`**

1. Open a terminal window and compile the code without any optimisation, for example
   **`gcc -o perfopt.exe perfopt.c`**
   and run it with
   **`perfopt.exe`** (Windows) or **`./perfopt.exe`** (Mac/Linux)
   Record the execution time that the code prints out.

2. Repeat step 1 with “standard” optimisation:
   **`gcc -O -o perfopt.exe perfopt.c`**

3. Repeat step 2 with “full” optimisation:
   **`gcc -O3 -o perfopt.exe perfopt.c`**

4. Now look at the code: can you say which of the loop nests (A, B, C and D) would be appropriate targets for optimisation?

5. Make a copy of the code (say as **`perfopt2.c`**) and edit it to swap the index order for the array **`x`** (i.e. replace all instances of **`x[j][i]`** with **`x[i][j]`**). Compile this version with full optimisation and run it. Can you explain the outcome? Consider whether this change might affect the readability/maintainability of the code.

6. Make another copy of the new code (say as **`perfopt3.c`**) and edit it as follows: separate out the j=0 case in loop nest B and the i=0 case in loop nest C. Now merge the two resulting double loop nests together into one. Compile this version with full optimisation and run it. Can you explain the outcome? Consider whether this change might affect the readability/maintainability of the code.